

PROTOCOL

TITLE: PHASE II, RANDOMIZED, MULTICENTER, DOUBLE-BLIND, PLACEBO-CONTROLLED TRIAL IN PATIENTS WITH OVARIAN CANCER

STUDY PRJ5457C

STUDY DRUG: CMP-135

IND: 892018

MEDICAL MONITOR:

SPONSOR: Company, Inc.

DATE AMENDED:

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Assessments or Procedures		Study Period ^a			
	Screening		Every 4	Every 8	

PROTOCOL SYNOPSIS

TITLE:

STUDY DRUG: CMP-135

INDICATION: Ovarian Cancer

IND:

SPONSOR: Company, Inc

DATE FINAL:

DATE AMENDED:

OBJECTIVES

Primary Objective

The primary objective for this study is as follows:

- To estimate the clinical benefit of the addition of CMP-135 as maintenance therapy for patients with ovarian cancer in a second or third complete remission, as measured by investigator-determined PFS by radiographic assessment

Secondary Objectives

Secondary objectives for this study are as follows:

- To make a preliminary assessment of the safety and tolerability of CMP-135 in this population
- To make a preliminary assessment of the efficacy of CMP-135 as measured by overall survival

STUDY DESIGN

The proposed study is a Phase II, randomized, placebo-controlled, double-blind, multicenter clinical trial of CMP-135 in patients with ovarian cancer in a second or third complete remission. Patients will be randomized in a 1:1 ratio to either CMP-135 or placebo. Randomization will be stratified based on whether their cancer is in a second or third complete remission. Patients will receive 150 mg of CMP-135 or placebo daily by mouth. Treatment will continue until evidence of radiographic progression, until the patient experiences intolerable toxicities most probably attributable to CMP-135, or until the patient withdraws from the study. Day 1 of the study will be defined as the first day the patient receives CMP-135 or placebo. Patients who discontinue study treatment (i.e., due to documented disease progression) will be followed for survival approximately every 3 months until death, loss to follow-up, or study termination by Company.

OUTCOME MEASURES

Primary Efficacy Outcome Measure

The primary outcome measure is PFS, defined as the time from randomization to the first occurrence of radiographic progression or death from any cause, determined by the investigator. Radiographic progression is defined as the appearance of a new lesion on a computed tomography (CT) scan of the chest/abdomen/pelvis consistent with recurrent disease, including new onset or worsening ascites. New ambiguous lesions may be biopsied to determine progression. The patient should remain on blinded study treatment (CMP-135 or placebo) until radiographic progression is confirmed. In the case of clinical deterioration, such as small bowel obstruction, a CT scan should be performed to determine whether radiographic progression has occurred. If radiographic progression cannot be documented objectively, the investigator must consult the Medical Monitor before removing a patient from the study. Increasing levels of CA-125 in the absence of radiographic progression has not been validated as an endpoint in this setting and should not be interpreted as progression or as a criterion for discontinuation of study treatment.

For patients who discontinue study treatment prior to documented radiographic progression, every effort should be made to continue scheduled tumor assessments so that the date of progression can be documented.

Secondary Efficacy Outcome Measures

Secondary efficacy outcome measures for this study include the following:

- Overall survival, defined from the time of randomization

Safety Outcome Measures

Safety outcome measures for this study include the following:

- The number and attribution of all adverse events (including vital signs, physical findings, and clinical laboratory results) that occur in patients who have received CMP-135 or placebo

SAFETY PLAN

The collection of all adverse events and serious adverse events will begin on the day of randomization and will continue for 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. Any serious adverse events that occur after informed consent but prior to randomization and that are deemed to be related to study-mandated procedures will be reported. All randomized patients will be monitored for adverse events regularly by Company staff via electronically captured data that allow for near real-time evaluation of reported events. In addition, this monitoring will include investigations of possible safety trends with respect to study drug. Serious adverse events are to be reported in an expedited manner within 24 hours to the Medical Monitor and to Company's Drug Safety Department.

Although no specific concomitant medications except for non-protocol-specified anti-tumor therapy are prohibited during this study, CMP-135 may potentially affect the pharmacokinetics of other drugs by altering their metabolism. In addition, metabolic inducers could possibly modify the pharmacokinetics of CMP-135. Therefore, concomitant medications should be used with care, and the risk-benefit profile of each agent should be taken into consideration. Appendix B lists concomitant medications that may have the potential to interact with CMP-135.

STUDY TREATMENT

Patients will receive 150 mg of CMP-135 daily by mouth or placebo daily by mouth, beginning on Day 1, continuously until radiographic disease progression, intolerable toxicity most probably attributable to CMP-135 or placebo, or withdrawal from the study.

Women of childbearing potential must have a negative pregnancy test on the day they receive their monthly supply of CMP-135 or placebo. The patient's use of two effective forms of contraception (including one barrier method) must be confirmed before dispensing CMP-135 or placebo.

CONCOMITANT THERAPY AND CLINICAL PRACTICE

Concurrent non-protocol-specified anti-tumor therapy (e.g., chemotherapy, hormonal therapy, other targeted therapy, investigational agents, radiation therapy, surgery) will not be allowed on this study. Although no specific non-anti-tumor concomitant medications are prohibited during this study, CMP-135 may potentially affect the pharmacokinetics of other drugs by altering their metabolism. In addition, metabolic inducers could possibly modify the pharmacokinetics of CMP-135. Therefore, concomitant medications should be used with care, and the risk-benefit profile of each agent should be taken into consideration. Appendix B lists concomitant medications that may have the potential to interact with CMP-135.

STATISTICAL METHODS

Primary Efficacy Analysis

The primary efficacy endpoint in this study is the hazard ratio (HR) of PFS using the stratified Cox model. PFS will be defined as the time from randomization until the first occurrence of radiographic progression or death from any cause, determined by the investigator (see Primary Efficacy Outcome Measures, Section 3.3.1, for a definition of radiographic progression). Patients who do not meet the criteria for disease progression, but who die within 45 days of last treatment dose, will be treated as having progressed for the PFS analysis. Patients who do not meet the criteria for disease progression and who do not die will be censored at the time of last tumor assessment.

The stratification factor for the stratified Cox model will be remission history (second remission/third remission). This primary analysis will include a 90% confidence interval (CI) for the stratified HR for PFS. Results from a stratified and unstratified log-rank test will also be presented. Kaplan-Meier methodology will also be used to estimate the PFS curves and medians in each treatment arm.

Missing Data

For PFS, patients who are lost to follow-up will be analyzed as censored observations on the last date that the patient was known to be progression free plus 1 day. If no post-randomization tumor assessment is available, the observation will be censored on the day after the randomization date. For duration of survival, patients who are lost to follow-up will be analyzed as censored observations on the date of last contact.

Determination of Sample Size

The sample size of approximately 219 patients for this study.

Interim Analysis

No interim analyses for efficacy are planned.

LIST OF ABBREVIATIONS AND DEFINITION OF TERMS

AE adverse event
BCC basal cell carcinoma
BUN blood urea nitrogen
CI confidence interval
CRA clinical research associate
CRO contract research organization
CT computed tomography
EC ethics committee
ECOG Eastern Cooperative Oncology Group
eCRF electronic Case Report Form
EDC electronic data capture
FDA Food and Drug Administration
GCP Good Clinical Practice
HDPE high-density polyethylene
HEENT head, eye, ear, nose, and throat
HR hazard ratio
ICH International Conference on Harmonisation
IHC immunohistochemistry
IND Investigational New Drug
IRB Institutional Review Board
ITT intent to treat
IV intravenous
IVRS interactive voice response system
IWRS interactive web response system
NCI CTCAE National Cancer Institute Common Terminology
Criteria for Adverse Events
NF National Formulary
PFS progression-free survival
PK pharmacokinetic
PS performance status
RBC red blood cell
RT-PCR reverse transcriptase polymerase chain reaction
SAE serious adverse event
SMO Smoothed (7-pass transmembrane domain protein)
ULN upper limit of normal
USP United States Pharmacopeia
WBC white blood cell

1. BACKGROUND

Ovarian cancer is a disease with ongoing, significant unmet medical need. In the United States, an estimated 21,650 new cases of ovarian cancer will be diagnosed, with an estimated 15,520 deaths in 2008 (American Cancer Society 2008). Primary peritoneal carcinoma and fallopian tube carcinoma are less commonly occurring cancers that are treated the same way as ovarian cancer. Most patients with ovarian cancer are diagnosed with advanced (Stage III or Stage IV) disease and are treated with surgery followed by either intravenous (IV) or intraperitoneal chemotherapy. This chemotherapy typically consists of a platinum agent (cisplatin or carboplatin) combined with a taxane (e.g., paclitaxel). Although many patients respond to initial therapy, the majority will experience disease recurrence and will require additional chemotherapy. Therefore, maintenance chemotherapy has been studied in the first-line setting as a potential way to decrease or delay the recurrence of ovarian cancer after initial surgery and chemotherapy.

2. OBJECTIVES

2.1 PRIMARY OBJECTIVE

The primary objective for this study is as follows:

- To estimate the clinical benefit of the addition of CMP-135 as maintenance therapy for patients with ovarian cancer in a second or third complete remission, as measured by investigator-determined PFS by radiographic assessment

2.2 SECONDARY OBJECTIVES

Secondary objectives for this study are as follows:

- To make a preliminary assessment of the safety and tolerability of CMP-135 in this population
- To make a preliminary assessment of the efficacy of CMP-135 as measured by overall survival

2.3 EXPLORATORY OBJECTIVES

3. STUDY DESIGN

3.1 DESCRIPTION OF THE STUDY

The proposed study is a Phase II, randomized, placebo-controlled, double-blind, multicenter clinical trial of CMP-135 in patients with ovarian cancer in a second or third complete remission. Patients will be randomized in a 1:1 ratio to either CMP-135 or placebo. Randomization will be stratified based on whether their cancer is in a second or third complete remission. Patients will receive 150 mg

of CMP-135 or placebo daily by mouth. Treatment will continue until evidence of radiographic progression, until the patient experiences intolerable toxicities most probably attributable to CMP-135, or until the patient withdraws from the study. Day 1 of the study will be defined as the first day the patient receives CMP-135 or placebo. Patients who discontinue study treatment (i.e., due to documented disease progression) will be followed for survival approximately every 3 months until death, loss to follow-up, or study termination by Company.

3.2 RATIONALE FOR STUDY DESIGN

This study is designed to compare CMP-135 with placebo as maintenance therapy for patients with ovarian cancer in a second or third complete remission to determine if CMP-135 can prolong PFS. The continuous dosing schedule of CMP-135 was chosen, based on the pharmacokinetic (PK) properties of CMP-135 characterized in the Phase I study of this agent, to maximize exposure and potential therapeutic effect. Along with assessments for the safety and tolerability of CMP-135 or placebo, patients will be evaluated for disease progression at predefined, standard intervals to minimize evaluation time biases.

3.3 OUTCOME MEASURES

3.3.1 Primary Efficacy Outcome Measure

The primary outcome measure is PFS, defined as the time from randomization to the first occurrence of radiographic progression or death from any cause, determined by the investigator.

Radiographic progression is defined as the appearance of a new lesion on a computed tomography (CT) scan of the chest/abdomen/pelvis consistent with recurrent disease, including new onset or worsening ascites. New ambiguous lesions may be biopsied to determine progression. The patient should remain on blinded study treatment (CMP-135 or placebo) until radiographic progression is confirmed. In the case of clinical deterioration, such as small bowel obstruction, a CT scan should be performed to determine whether radiographic progression has occurred. If radiographic progression cannot be documented objectively, the investigator must consult the Medical Monitor before removing a patient from the study. Increasing levels of CA-125 in the absence of radiographic progression has not been validated as an endpoint in this setting and should not be interpreted as progression or as a criterion for discontinuation of study treatment.

For patients who discontinue study treatment prior to documented radiographic progression, every effort should be made to continue scheduled tumor assessments so that the date of progression can be documented.

3.3.2 Secondary Efficacy Outcome Measures

Secondary efficacy outcome measures for this study include the following:

- Overall survival, defined from the time of randomization

3.3.3 Safety Outcome Measures

Safety outcome measures for this study include the following:

- The number and attribution of all adverse events (including vital signs, physical findings, and clinical laboratory results) that occur in patients who have received CMP-135 or placebo

3.3.4 Exploratory Outcome Measures

3.4 SAFETY PLAN

The collection of all adverse events and serious adverse events will begin on the day of randomization and will continue for 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. Any serious adverse events that occur after informed consent but prior to randomization and that are deemed to be related to study-mandated procedures will be reported. All randomized patients will be monitored for adverse events regularly by Company staff via electronically captured data that allow for near real-time evaluation of reported events. In addition, this monitoring will include investigations of possible safety trends with respect to study drug. Serious adverse events are to be reported in an expedited manner within 24 hours to the Medical Monitor and to Company's Drug Safety Department.

3.5 STUDY PATIENTS

The study population of women with ovarian cancer in second or third complete remission was chosen.

3.6 CONTROL GROUPS

There is no approved or generally recognized standard of care maintenance therapy for ovarian cancer in second or third complete remission, and there is sparse historical data relating to outcomes in this setting. Because of this, a double-blinded placebo control will be used in this randomized study.

3.7 MINIMIZATION OF BIAS

Patients will be allocated to treatment arms through the use of a stratified randomization algorithm that will balance the proportion of patients in each stratum. The stratification factor (whether the patient's cancer is in a second or third complete remission) was chosen because of its potential to affect efficacy and/or safety outcomes, thus minimizing differences in treatment arms due to sources other than CMP-135. The appearance of placebo pills will be indistinguishable from CMP-135 pills and will have identical shape and color. Additionally, the pill bottles for CMP-135 will look identical to those for placebo, with only the differences in bottle numbers to distinguish them.

As the result of the size of the study, the randomization, blinding, and expected low incidence of unblinding adverse events, collected data will be protected from biases that could arise through subjectivity in the reporting of outcome measures.

All laboratory tests of blood specimens will be performed by local laboratories.

3.8 ADMINISTRATIVE STRUCTURE

This study will be sponsored by Company, Inc. A total of approximately 100 patients will be enrolled at approximately 50 centers in approximately 10 countries. All patients will be enrolled in a single stage. There will be no limit on the number of patients enrolled in each site. Provided that safety concerns do not arise, there will be no interruptions in enrollment.

Outsourcing services will include a contract research organization (CRO); Interactive Voice Response System (IVRS) and Interactive Web Response System (IWRS); monitoring; drug depot; central Institutional Review Board (IRB); and central laboratory vendors. The set up and management of these vendors will be managed by Company or a CRO.

3.9 COMPLIANCE WITH LAWS AND REGULATIONS

This study will be conducted in accordance with the U.S. Food and Drug Administration (FDA) regulations, the International Conference on Harmonisation (ICH) E6 Guideline for Good Clinical Practice (GCP), the Declaration of Helsinki (October 1996) and applicable local, state, and federal laws, as well as other applicable country laws.

4. MATERIALS AND METHODS

4.1 PATIENTS

4.1.2 Inclusion Criteria

Patients must meet the following inclusion criteria to be eligible for study entry:

- Age ≥ 18 years
- Histologic diagnosis of epithelial ovarian carcinoma, primary peritoneal carcinoma, or fallopian tube carcinoma
- Must be in second or third complete remission, have received chemotherapy (platinum-based and/or non-platinum-based) for recurrent disease, and have achieved a complete remission after their most recent chemotherapy regimen

Complete remission is defined as no symptoms suggestive of persistent cancer, CT scan of the chest/abdomen/pelvis without evidence of ovarian cancer within 4 weeks of randomization, and normal CA-125 (measured within 2 weeks of randomization) following completion of prior chemotherapy.

The study investigator should confirm the status of disease remission by CT scan before patient enrollment.

If patient has lymphadenopathy by CT scan and the investigator thinks that it is unlikely due to ovarian cancer, this patient is considered eligible. If indicated, a confirmatory biopsy should be performed.

- Patients must have completed their most recent dose of cytotoxic chemotherapy regimen (platinum-based or non-platinum based) no less than 3 weeks and no more than 14 weeks prior to randomization.
 - Archival tissue must be available and requested.
- The tissue must consist of representative tumor specimens in paraffin blocks (preferred) or at least 15 unstained slides, with an associated pathology report, obtained at any time prior to entry of study. Archival tissues are to be sent as soon as possible.
- Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1 (see Appendix D)
- Adequate hematopoietic capacity, as defined by the following:

Hemoglobin > 8.5 g/dL, not transfusion-dependent Granulocyte count $\geq 1200/\mu\text{L}$ Platelets $\geq 75,000/\mu\text{L}$

- For women of childbearing potential: Use of two effective methods of barrier contraception, including one barrier method (See Appendix C for acceptable and unacceptable methods of contraception.)
- Signed informed consent

4.1.3 Exclusion Criteria

Patients who meet any of the following criteria will be excluded from study entry:

- Pregnancy or lactation

For women of childbearing potential: use of two effective methods of barrier contraception, including one

barrier method, during the study and for 12 months after discontinuing study drug (see Appendix C for acceptable and unacceptable methods of contraception).

- Patients whose ovarian cancer is in first remission
- Patients must not have experienced more than two prior recurrences of disease
- Uncontrolled medical illnesses such as infection requiring IV antibiotics
- Life expectancy < 12 weeks

4.2 METHOD OF TREATMENT ASSIGNMENT AND BLINDING

After written informed consent has been obtained and eligibility has been established (with archival tissue identified and available for research testing), the study site will obtain the patient's identification number and treatment assignment from the IVRS and IWRS. Patients may be randomized up to 5 calendar days prior to receiving their first dose of treatment (Day 1).

The investigator and the patient will be blinded to treatment assignment. The unblinding of a patient's treatment assignment (to the patient and the investigator) prior to the final study analysis is permitted only in the event of a serious study treatment-related toxicity (as part of the Investigational New Drug [IND] safety reporting process). All cases of safety unblinding require the approval of the Medical Monitor (see Section 5.4 for Medical Monitor contact information).

4.3 STUDY TREATMENT

4.3.1 Formulation

The formulated drug product is provided in hard gelatin capsules of 150-mg strength. The capsules are Size 1 with a grey-colored cap and a pink-colored body. Placebo will consist of excipients for CMP-135 without the active molecule. Placebo will be supplied by Company in hard gelatin capsules matching the active drug product in color and size.

Active and placebo capsules will be packaged in 75-mL round, white, high-density polyethylene (HDPE) bottles and closed with two-piece HDPE child-resistant caps. The bottles will be induction-sealed and labeled for clinical use. Each bottle will contain 32 capsules of a given strength, enough for 4 consecutive weeks of dosing with overage. For further details, see the CMP-135 Investigator Brochure.

Active and placebo capsules will be packaged in blind-labeled bottles.

4.3.2 Dosage, Administration, and Storage

Patients will receive 150 mg of CMP-135 daily by mouth or placebo daily by mouth, beginning on Day 1, continuously until one of the following occurs:

- Radiographic disease progression
- Intolerable toxicity most probably attributable to CMP-135 or placebo
- Withdrawal from the study

Women of childbearing potential must have a negative pregnancy test on the day they receive their monthly supply of CMP-135 or placebo. The patient's use of two effective forms of contraception (including one barrier method) must be confirmed before dispensing CMP-135 or placebo.

Study treatment should be taken at approximately the same time each day, with or without food. If a patient misses a dose (e.g., because of emesis), the patient should be instructed not to take or make up that dose and to resume dosing with the next scheduled dose. Missed doses will not be made up with the next dose. If a patient forgets a dose at their regular time period and remembers later on that day, patient may still take the dose that same day. Patients will be instructed to bring all unused capsules and their medication diary to each study visit for assessment of compliance.

The terminal half-life of CMP-135 has been determined to be 10–14 days in humans based on preliminary findings in healthy subjects. If plasma levels of CMP-135 need to be lowered emergently, animal studies suggest that oral administration of activated charcoal may lower drug plasma levels more quickly than dose cessation alone.

4.3.3 Dosage Modification

Treatment with CMP-135 or placebo may be interrupted for up to 4 weeks for evaluation of an intolerable toxicity finding or up to 8 weeks for a planned surgical procedure. The Company Medical Monitor must be consulted in cases where study treatment will be interrupted.

Intolerable toxicities are defined as new (not present at baseline) Grade 3 or 4 adverse events considered related to CMP-135 that are potentially life-threatening or irreversible, and when, in the opinion of the investigator, the risk outweighs the benefit of continued treatment with CMP-135. The following adverse events are not considered intolerable:

- Grade 3 or 4 events that in the opinion of the investigator are more likely related to ongoing or recent procedures or concomitant medications other than CMP-135
- Hematologic or metabolic/chemistry laboratory abnormalities that are found on routine testing and are not considered clinically significant
- Musculoskeletal abnormalities, skin ulceration, fracture, debridement, or wound care
- Nausea, vomiting, or diarrhea that are adequately controlled after optimization of medical management.
- Grade 3 infection that is transient and treatable or manageable
- Asymptomatic or mild symptomatic (e.g., leg pain) thromboembolism found incidentally on imaging and managed with symptomatic control and anti-coagulation therapy
- Patients with an asymptomatic or tolerable severe adverse event may continue to receive study treatment provided that the adverse event is manageable and the patient and the investigator agree that continued study participation is acceptable.

If a treatment interruption occurs and it is determined that CMP-135 or placebo will be restarted, the original dose will be maintained. No dose reductions of CMP-135 or placebo are planned.

If a patient is suspected to be pregnant, CMP-135/placebo must be immediately discontinued. If it is later confirmed that the patient is not pregnant, dosing may be resumed.

4.4 CONCOMITANT AND EXCLUDED THERAPIES

Concurrent non-protocol-specified anti-tumor therapy (e.g., chemotherapy, hormonal therapy, other targeted therapy, investigational agents, radiation therapy, surgery) will not be allowed on this study (see Exclusion Criteria, Section 4.1.3).

Although no specific non-anti-tumor concomitant medications are prohibited during this study, CMP-135 may potentially affect the pharmacokinetics of other drugs by altering their metabolism. In addition, metabolic inducers could possibly modify the pharmacokinetics of CMP-135. Therefore, concomitant medications should be used with care, and the risk-benefit profile of each agent should be taken into consideration. Appendix B lists concomitant medications that may have the potential to interact with CMP-135.

4.5 STUDY ASSESSMENTS

The study period will begin on Day 1, and patients will be followed until documented disease progression, death, or withdrawal of consent from study.

For this study, all adverse events and serious adverse events will be collected for all patients beginning from the day of randomization to 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. Refer to Section 5.0 for further details regarding the collection of adverse events.

Concomitant medications will be collected for all randomized patients beginning from Day – 14 to the end of the study period, beginning at informed consent. Record all concomitant medications administered within the 14 days preceding Day 1.

4.5.1 Definitions of Study Assessments

a. Vital Signs

Vital signs will include measurements of pulse, systolic and diastolic blood pressure while the patient is in a seated position, respiratory rate, and temperature.

b. Physical Examination

A complete physical examination should include the evaluation of head, eye, ear, nose, and throat (HEENT), cardiovascular, dermatological, musculoskeletal, respiratory, gastrointestinal, gynecologic, and neurological systems. Changes from baseline abnormalities should be recorded at each subsequent physical examination. New or worsened abnormalities should be recorded as adverse events if appropriate.

c. Tumor and Response Evaluation

Tumor assessments will be performed at screening and every 8 weeks during the study. CT scans should include chest, abdomen, and pelvis scans. At the investigator's discretion, CT scans may be repeated at any time if progressive disease is suspected. If CT scans are obtained to assess disease progression when the patient has elevated CA-125 or is symptomatic and there is no radiographic evidence of disease progression, subsequent CT scan should be performed 8 weeks from that last CT scan. CT scan evidence of progressive disease is to be confirmed by study investigator prior to taking patients off study.

CA-125 levels will also be monitored in patients with a history of tumors that secrete this serum marker. See Laboratory Assessments for additional details.

d. Laboratory Assessments

Samples for hematology, serum chemistries, urinalysis, pregnancy test, and CA-125 will be analyzed at the study site's local laboratory. Laboratory assessments will include the following:

- Hematology (hemoglobin, hematocrit, platelet count, red blood cell [RBC] count, white blood cell [WBC] count, and percent or absolute differential count [neutrophils, eosinophils, lymphocytes, monocytes, basophils, and other cells])
- Serum chemistries (including glucose, blood urea nitrogen [BUN], creatinine, sodium, potassium, magnesium, bicarbonate, calcium, total protein, albumin, total bilirubin, alkaline phosphatase, AST, and ALT)
- Urinalysis (pH, specific gravity, glucose, protein, ketones, and blood)

- Pregnancy test: All women of childbearing potential (see Appendix C for definition, which includes women who have had a tubal ligation) will have a serum pregnancy test at screening. Urine pregnancy tests will be performed every 4 weeks for women of childbearing potential. If a urine pregnancy test is positive, it must be confirmed by a serum pregnancy test.
- Serum CA-125

Patients who do not have a history of tumors that secrete CA-125 are required to have a CA-125 measurement only at screening.

For patients with a history of CA-125-secreting tumors (or if the CA-125-secretion status of the patient's tumor is unknown), CA-125 levels will be measured at each tumor assessment. If a patient's CA-125 is $\geq 2 \times$ ULN at a tumor assessment timepoint, a repeat CA-125 level and a CT scan of the chest/abdomen/pelvis may be obtained 4 weeks later, but are optional.. The patient will remain on study treatment until radiographic disease progression is confirmed. During the period of elevated CA-125 level $\geq 2 \times$ ULN and before evidence of radiographic progression, CA-125 measurements may be performed every 4 weeks. Medical Monitor should be consulted if the patient is symptomatic, but without evidence of radiographic progression.

- Archival tumor tissue

Representative tumor specimens in paraffin blocks (strongly preferred) or at least 15 unstained slides, with an associated pathology report, must be available and requested at any time prior to randomization. Archival tissues are to be sent as soon as possible.

4.5.2 Screening and Pretreatment Assessments (Days – 14 to – 1)

Screening tests and evaluations will be used to determine the eligibility of each patient for study inclusion. All patients must provide written informed consent before any study-specific procedures and assessments are performed and any exclusionary medications are discontinued for the purposes of establishing eligibility. After the patient signs informed consent, the study staff will obtain the patient's identification (screening) number from the IVRS. After establishing the patient's eligibility, the site will obtain patient's randomization (subject) number from the IVRS.

Screening evaluations will be performed within the 14 days preceding Day 1, with the exception of the baseline tumor assessment, which may be performed within 28 days prior to randomization. Results of tests or examinations

performed as standard of care prior to obtaining informed consent may be used rather than repeating required tests.

Concomitant medications taken within 14 days preceding Day 1 will be recorded and concomitant medication recording will continue through the end of the study period.

Any serious adverse events that occur after informed consent but prior to randomization and which are deemed to be related to study-mandated procedures should be reported (see Section 5.2.1).

Please see the Study Flowchart provided in Appendix A for the schedule of screening and pretreatment assessments.

4.5.3 Assessments during Treatment

Once a patient has signed the Informed Consent Form, meets all inclusion and exclusion criteria and is randomized into the study, the patient will be scheduled for Day 1 assessments. Patients will receive 150 mg of CMP-135 daily by mouth or placebo daily by mouth, beginning on Day 1.

A window of ± 3 days will be allowed for all visits and radiographic assessments performed during the study period. At the discretion of the study investigator, the visit and radiographic assessment window may be increased up to 1 week to accommodate patient scheduling (e.g., travel, vacation, holiday). For laboratory assessments, a window of up to 4 days prior to a scheduled treatment day will be allowed. Study medications will only be dispensed after the scheduled clinical, radiographic, and laboratory assessments are confirmed by the study investigator and the investigator believes that the patient should continue study drug/placebo. If a visit window has been delayed (maximum 1 week), an adequate CMP-135/placebo drug supply should be dispensed to accommodate the delay until the next visit.

Please see the Study Flowchart provided in Appendix A for the schedule of treatment period assessments.

4.5.4 Treatment Discontinuation

Please see the Study Flowchart provided in Appendix A for assessment to be performed at the time of treatment discontinuation. Tumor assessments should continue for patients who discontinue treatment for reasons other than documented disease progression or patient withdrawal of consent.

4.5.5 Survival Follow-Up Assessments

Patients who discontinue study treatment after documented disease progression will be followed for survival approximately every 3 months ($\pm$ 1month) until death, loss to follow-up, physician decision to withdraw the patient from the study, the patient's decision to withdraw from the study, or study termination by Company. Patients will be followed up for safety until 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. Follow-up visits will consist of at least a telephone contact with the patient, a family member, or the physician to inquire about the patient's survival status.

4.6 PATIENT DISCONTINUATION

The investigator has the right to discontinue a patient from the study a) for any medical condition that the investigator determines may jeopardize the patient's safety if she continues in the study; b) for reasons of noncompliance (e.g., missed doses, visits); c) if the patient becomes pregnant; or d) if the investigator determines it is in the best interest of the patient.

See Section 4.5.4 for assessments that are to be performed for patients who prematurely withdraw from the study during the treatment period.

4.7 STUDY DISCONTINUATION

Company has the right to terminate this study at any time. Reasons for terminating the study may include, but are not limited to, the following:

- The incidence or severity of adverse events in this or other studies indicates a potential health hazard to patients.
- Patient enrollment is unsatisfactory.
- Data recording is inaccurate or incomplete.

4.8 ASSAY METHODS

Submission of a paraffin-embedded tumor specimen (blocks are strongly preferred) or unstained paraffin slides (minimum of 15 slides) of representative tumor will be required for all patients who are enrolled in the study.

4.9 STATISTICAL METHODS

This study is a randomized, placebo-controlled trial of approximately 100 patients designed to make a preliminary assessment of the safety and efficacy of CMP-135 as maintenance therapy in patients with ovarian cancer in second or third complete remission. The focus of the trial is to estimate the specified treatment effects in addition to characterizing safety. In addition to the primary and secondary analyses described below, descriptive summaries of continuous data will include the mean, standard deviation, median, minimum, maximum, and number of patients. Descriptive summaries of discrete data will include the number of patients and incidence as a frequency and percentage.

4.9.1 Analysis of the Conduct of the Study

Enrollment, study treatment administration, and discontinuations from the study will be summarized. The incidence of treatment discontinuation will be tabulated.

4.9.2 Analysis of Treatment Group Comparability

Demographic variables, stratification factors, and other baseline characteristics will be summarized.

4.9.3 Efficacy Analyses

Unless otherwise noted, all efficacy analyses will be performed using the intent-to-treat (ITT) patient population defined as all randomized patients, and patients will be allocated to the treatment arm to which they were randomized.

a. Primary Endpoint

The primary efficacy endpoint in this study is the hazard ratio (HR) of PFS using the stratified Cox model. PFS will be defined as the time from randomization until the first occurrence of radiographic progression or death from any cause, determined by the investigator (see Primary Efficacy Outcome Measures, Section 3.3.1, for a definition of radiographic progression). Patients who do not meet the criteria for disease progression, but who die within 45 days of last treatment dose, will be treated as having progressed for the PFS analysis. Patients who do not meet the criteria for disease progression and who do not die will be censored at the time of last tumor assessment.

The stratification factor for the stratified Cox model will be remission history (second remission/third remission). This primary analysis will include a 90% confidence interval (CI) for the stratified HR for PFS. Results from a stratified and unstratified log-rank test will also be presented. Kaplan-Meier methodology will also be used to estimate the PFS curves and medians in each treatment arm.

b. Secondary Endpoints

Secondary endpoints include PFS for patients with Cnc ligand-overexpressing tumors at baseline and overall survival.

The HR of PFS using the stratified Cox model in the primary analysis will be repeated with the addition of Cnc ligand expression as a covariate. Cnc ligand expression in archival tissue by IHC will be quantified as either a continuous or semi-quantitative measurement. Cnc ligand expression will be included in the Cox model as either a continuous or ordinal variable depending on the occurrence of apparent or meaningful ordinal groupings. The stratified log-rank test may also be repeated using Cnc ligand expression as a stratification factor.

Duration of overall survival is defined as the time from randomization until death by any cause. Analyses performed for the primary endpoint will be repeated using overall survival. Additionally, Cnc ligand expression may be included in the stratified Cox model as well as the stratified log-rank test.

4.9.4 Safety Analyses

Safety analyses will include all patients who receive at least one dose of study treatment. Patients who receive at least some amount of CMP-135 will be allocated to the CMP-135 treatment arm. Safety will be assessed through summaries of all adverse events in this population, including serious adverse events and adverse events leading to discontinuation of CMP-135 or placebo. All treatment-emergent adverse events will be mapped to thesaurus terms and graded by National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v3.0.

4.9.5 Exploratory Analyses

The effect of Cnc expression and/or pathway signaling on primary and secondary endpoints may be examined. Measures of expression or pathway signaling may include, but will not be limited to, one or more of the following:

- 1 Assessments related to the understanding of CMP-135's mechanism of action and activity, ovarian cancer pathogenesis, and/or ovarian cancer treatment.

In addition, as an exploratory analysis, PFS will also be analyzed as a combination of radiographic and CA-125 criteria. Patients whose tumors do not secrete CA-125 will not be evaluable for this analysis.

4.9.6 Missing Data

For PFS, patients who are lost to follow-up will be analyzed as censored observations on the last date that the patient was known to be progression free plus 1 day. If no post-randomization tumor assessment is available, the observation will be censored on the day after the randomization date. For duration of survival, patients who are lost to follow-up will be analyzed as censored observations on the date of last contact.

4.9.7 Determination of Sample Size

The sample size of approximately 218 patients for this study.

4.9.8 Interim Analysis

No interim analyses for efficacy are planned. Patients will be monitored for safety as described in the Safety Plan, Section 3.4.

4.10 DATA QUALITY ASSURANCE

The data will be collected via Electronic Data Capture (EDC) using eCRFs. The site will be responsible for data entry into the EDC system. In the event of discrepant data, the CRO will request data clarification from the sites, which the sites will resolve electronically in the EDC system. The CRO will be responsible for the data management of this trial, including quality checking of the data.

Company will perform oversight of the data management of this trial. Company will produce an EDC Study Specification document that describes the quality checking to be performed on the data. Central Laboratory data and IVRS will be sent directly to Company, using Company's standard procedures to handle and process the electronic transfer of these data. eCRFs and correction documentation will be maintained in the EDC system's audit trail. System backups for data stored at Company and records retention for the study data will be consistent with Company's standard procedures.

5. ASSESSMENT OF SAFETY

5.1 SAFETY PARAMETERS AND DEFINITIONS

Safety assessments will consist of monitoring and recording protocol-defined adverse events (AEs) and serious adverse events (SAEs); measurement of protocol-specified hematology, clinical chemistry, and urinalysis variables; measurement of protocol-specified vital signs; and other protocol-specified tests that are deemed critical to the safety evaluation of the study drug(s).

Company or its designee is responsible for reporting relevant SAEs to the Competent Authority, other applicable regulatory authorities, and participating investigators, in accordance with ICH guidelines, FDA regulations, European Clinical Trials Directive, and/or local regulatory requirements.

Company or its designee is responsible for reporting unexpected fatal or life-threatening events associated with the use of the study drug to the regulatory agencies and competent authorities by telephone or fax within 7 calendar days after being notified of the event.

Company or its designee will report other relevant SAEs associated with the use of the study medication to the appropriate competent authorities (according to local guidelines), investigators, and central IRBs/ECs (except in the United States where investigators are responsible for reporting to their IRBs per local requirements) by a written safety report within 15 calendar days of notification.

5.1.1 Adverse Event

An AE is any unfavorable and unintended sign, symptom, or disease temporally associated with the use of an investigational medicinal product or other protocol-imposed intervention, regardless of attribution.

This includes the following:

- AEs not previously observed in the patient that emerge during the protocol-specified AE reporting period, including signs or symptoms associated with ovarian cancer that were not present prior to the AE reporting period (see Section 5.2.1)
- Complications that occur as a result of protocol-mandated interventions (e.g., invasive procedures such as biopsies)

- AEs that occur prior to assignment of study treatment that are related to a protocol-mandated intervention (e.g. invasive procedures such as biopsies, medication washout, or no treatment run-in).
- Preexisting medical conditions (other than the condition being studied), judged by the investigator to have worsened in severity or frequency or changed in character during the protocol-specified AE reporting period

5.1.2 Serious Adverse Event

An SAE is any AE that is any of the following:

- Fatal (i.e., the AE actually causes or leads to death)
- Life threatening (i.e., the AE, in the view of the investigator, places the patient at immediate risk of death)
- Requires or prolongs inpatient hospitalization
- Results in persistent or significant disability/incapacity (i.e., the AE results in substantial disruption of the patient's ability to conduct normal life functions)
- A congenital anomaly/birth defect in a neonate/infant born to a mother exposed to the investigational product(s)
- Considered a significant medical event by the investigator (e.g., may jeopardize the patient or may require medical/surgical intervention to prevent one of the outcomes listed above)

All AEs that do not meet any of the criteria for serious should be regarded as **non-serious AEs**.

The terms "severe" and "serious" are not synonymous. Severity refers to the intensity of an AE (as in mild, moderate, or severe pain); the event itself may be of relatively minor medical significance (such as severe headache). "Serious" is a regulatory definition and is based on patient or event outcome or action criteria usually associated with events that pose a threat to a patient's life or vital functions. Seriousness (not severity) serves as the guide for defining regulatory reporting obligations.

Severity and seriousness should be independently assessed when recording AEs and SAEs on the eCRF.

1.3 Protocol-Defined Events of Special Interest (Pregnancy, Abortion, and Congenital Anomaly/Birth Defect)

The following events are events of special interest and will need to be reported to the Sponsor expeditiously (see Section 5.4 for reporting instructions), irrespective of regulatory seriousness criteria:

a. Pregnancy

If a female patient becomes pregnant while receiving investigational therapy with CMP-135 or within 12 months after the last dose, a Pregnancy Report eCRF page should be completed, printed out, and submitted via fax with the Pregnancy Fax Coversheet expeditiously to the Sponsor (see Section 5.4.2) within 24 hours of learning of the pregnancy to facilitate outcome follow-up.

In the event the EDC system is unavailable, a paper Pregnancy Report form and Pregnancy Fax Coversheet should be completed and faxed to Company's Drug Safety Department or its designee at the fax numbers listed in Section 5.4.2.

b. Abortion, Congenital Anomaly, and Birth Defects

Abortions, congenital anomalies, and birth defects are events of special interest and will need to be reported to the Sponsor expeditiously (see Section 5.4 for reporting instructions).

Abortion, whether accidental, therapeutic, or spontaneous, should always be classified as serious (as the Sponsor considers these medically significant), recorded on an SAE eCRF page, and expeditiously reported to the Sponsor.

Any congenital anomaly/birth defect in a child born to a female patient exposed to the investigational product should be recorded and reported as an SAE.

5.2 METHODS AND TIMING FOR CAPTURING AND ASSESSING SAFETY PARAMETERS

The investigator is responsible for ensuring that all AEs and SAEs (as defined in Section 5.1) are recorded on the eCRF and reported to the Sponsor in accordance with protocol instructions.

5.2.1 Adverse Event Reporting Period

After informed consent, but prior to initiation of study medications, only SAEs caused by a protocol-mandated intervention will be collected (e.g., SAEs related to invasive procedures such as biopsies, medication washout, or no treatment run-in).

After initiation of study medications (CMP-135 or placebo), all AEs and SAEs regardless of attribution will be collected from the day of randomization until 45 days after the last dose of study treatment or after the initiation of new anti-tumor therapy, whichever is earlier. After this period, investigators should report only SAEs that are felt to be related to prior study treatment (see Section 5.6) and any pregnancy events that occur up to 12 months after the last dose of CMP-135 or placebo (see Section 5.1.3).

5.2.2 Eliciting Adverse Events

A consistent methodology of non-directive questioning for eliciting AEs at all patient evaluation time points should be adopted. Examples of non-directive questions include:

- “How have you felt since your last clinic visit?”
- “Have you had any new or changed health problems since you were last here?”

5.2.3 Assessment of Severity and Causality of Adverse Events

Investigators will seek information on AEs and SAEs at each patient contact. All AEs and SAEs, whether reported by the patient or noted by authorized study personnel, will be recorded in the patient’s medical record and on the appropriate AE/SAE eCRF page.

For each AE and SAE recorded on the applicable eCRF, the investigator will make an assessment of seriousness (see Section 5.1.2 for seriousness criteria), severity and causality.

Table 1 provides guidance for grading AE severity and Table Y provides guidance for assessing the causal relationship to the investigational product(s).

The AE grading (severity) scale found in the National Cancer Institute Common Terminology Criteria for Adverse Events (NCI CTCAE) v3.0 will be used for AE grading.

Table 1
Adverse Event Grading (Severity) Scale

Grade	Severity	Alternate Description ^a
1	Mild (apply event-specific NCI CTCAE grading criteria)	Transient or mild discomfort (< 48 hours); no interference with the patient's daily activities; no medical intervention/therapy required
2	Moderate (apply event-specific NCI CTCAE grading criteria)	Mild to moderate interference with the patient's daily activities; no or minimal medical intervention/therapy required
3	Severe (apply event-specific NCI CTCAE grading criteria)	Considerable interference with the patient's daily activities; medical intervention/therapy required; hospitalization possible
4	Very severe, life threatening, or disabling (apply event-specific NCI CTCAE grading criteria)	Extreme limitation in activity; significant medical intervention/therapy required, hospitalization probable
5	Death related to AE	

The NCI CTCAE v3.0 can be found: http://ctep.cancer.gov/reporting/ctc_v30.html
Note: Regardless of severity, some events may also meet regulatory serious criteria. Refer to definitions of an SAE (see Section 5.1.2). ^a Use these alternative definitions for Grade 1, 2, 3, and 4 events when the observed or reported AE is not in the NCI CTCAE listing.